

2025-2026 Spring

EE 374 Fundamentals of Power Systems

Term Project Definition

1. Context

Powerlab Co. is currently soliciting proposals for the development of a tailored software for distribution-level operations. While high-voltage transmission grids benefit from ubiquitous SCADA integration and automated metering infrastructure (AMI), the downstream distribution feeders remains largely "blind" during localized field maneuvers. Your firm is tasked with bridging this information gap by providing a tool that transforms static network data into dynamic operational insights.

Unlike transmission networks which may undergo frequent topological reconfigurations, the distribution segments serviced by our field engineers possess a fixed, well-defined topology. The connectivity matrices, line impedances, and manufacturer-specific equipment ratings are known beforehand. A standard data format and example files will be provided. Note that your product will also be tested with other data that has the same format.

However, the real-time "state" of the network—the instantaneous load profiles and capacitor bank statuses—is not telemetered. Consequently, the proposed software must feature an interface allowing engineers to manually input or edit current operating conditions. The engine must then map these dynamic variables onto the fixed physical infrastructure to resolve the system state.

The core of the suite must be a robust computational engine optimized for radial topologies. By exploiting the radial nature of these feeders, your firms should implement a solver that ensures near-instantaneous convergence. The primary objective is to provide field personnel with immediate visibility into voltage magnitudes and angles of all buses in the radial network. The tool should also highlight buses with voltage magnitude violations (overvoltage or undervoltage).

2. Objective and Requirements

Your firm is tasked with delivering a functional "product" rather than a mere script. The technical core must adhere to the following:

- i. Both backend and frontend of the application must be developed in Python environment.
- ii. The solver must utilize VA approach to calculate the voltage magnitude and angle at each bus of a radial three-phase balanced distribution network.
- iii. The tool must reliably process networks up to 10 buses.
- iv. Parse fixed network topology from a standardized file. A readme file that explains the data format will be provided.
- v. Visualize the network single-line-diagram. Indicating equipment such that the field engineers can understand and distinguish each. An example is given in Figure 1.

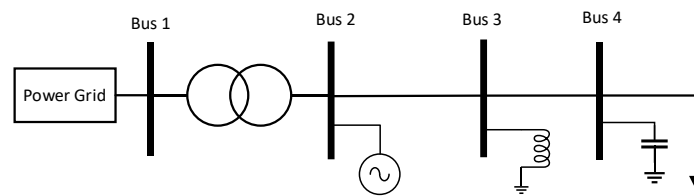


Figure 1 Example Network Visual

- vi. The interface must allow a field engineer to modify equipment data on-the-fly. The bare minimum of data that should be modifiable are as follows.
 - Load “active/reactive power” value.
 - Generator “active/reactive power” value.
 - Compensation unit “in service” value.
 - The voltage magnitude for the leaf node (i.e. Bus 4 in Figure 1).
- vii. The software must automatically highlight buses with voltage magnitude violations.
- viii. The final product must be a single executable file (.exe) compatible with Windows 11. It must be "portable"—meaning it runs out-of-the-box without requiring the user to install Python environments.

Firms are expected to architect a bespoke User Experience (UX) tailored specifically to the operational constraints of a field engineer. For example, your tool should be able to deal with false data type entries (non-numeric value to a numeric field).

3. Submission

Your submission will consist of four components. You are to put all four files in a .zip folder and submit it as such.

- **The Executable:** Submit the compiled application.
- **The Source Code:** Submit your source codes as a .pdf file.
- **User Manual:** A document (at most 7 pages) that is written in technical prose suitable for field engineers. Avoid overly academic jargon; instead, focus on operational instructions, input data formatting, and interpretation of results. The manual should follow the formatting rules below. A template will be provided.
 - **Line Spacing:** 1.15
 - **Margins:** 2 cm each side
 - **Fontsize:** 11 pt
 - **Font:** Times New Roman
 - **Text Alignment:** Justified
- **Infographic Page:** Submit an infographic page that relays information about the performance and use case of the application. This is a single-page document that advertises your product, visualizing attributes and technical specs.

4. Evaluation Criteria

Computational Accuracy	40%
UI/UX Design	30%
Professionalism & Customer Satisfaction	30%